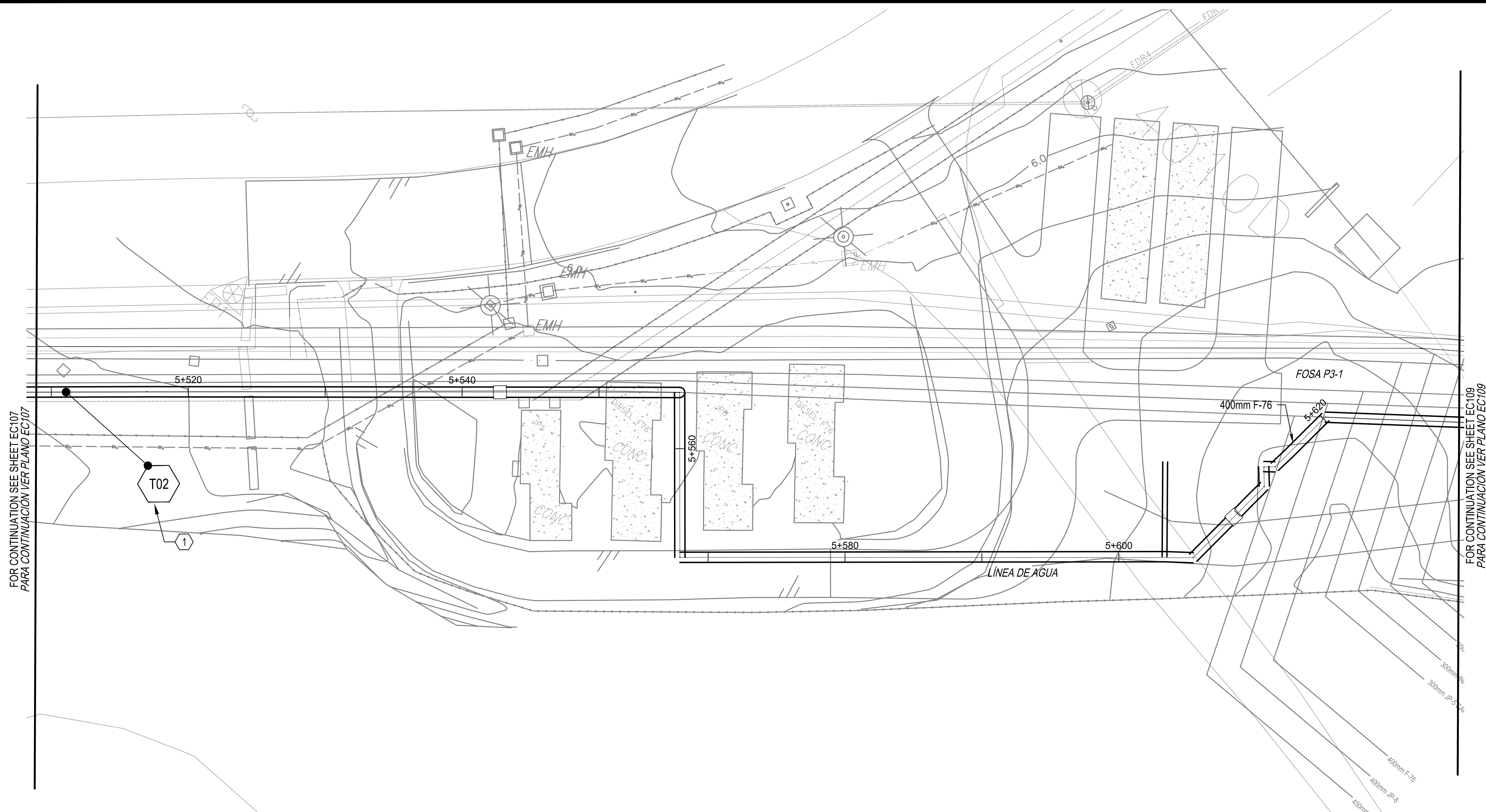


[illegible]



GENERAL NOTES:
NOTAS GENERALES:

1. EACH TEST STATION HAS 0 ANODES CONNECTED.
1. CADA ESTACIÓN DE PRUEBA TIENE 0 ÁNODOS CONECTADOS.

(B1)

CATHODIC PROTECTION PLAN,
PLANTA DE PROTECCIÓN CATÓDICA

1:200

KEYED NOTES:
NOTAS DE REFERENCIA:

1. CATHODIC PROTECTION TEST STATION
1. *ESTACIÓN DE PRUEBAS DE PROTECCIÓN CATÓDICA.*
2. INSULATED FLANGE KIT (IFK)
2. *KIT DE BRIDA AISLADA (IFK)*

[illegible]

FILE NAME: U:\231423\04-Civil\01-SHEETS\UTIL-1606884-E-EC110.dwg LAYOUT NAME: EC110 PLOTTED: Tuesday, June 25, 2024 - 2:10pm USER: 6450



GENERAL NOTES:
NOTAS GENERALES:

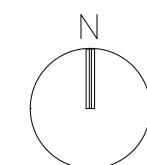
1. EACH TEST STATION HAS 0 ANODES CONNECTED.
1. CADA ESTACIÓN DE PRUEBA TIENE 0 ANODOS CONECTADOS.

B1

CATHODIC PROTECTION PLAN
PLANTA DE PROTECCIÓN CATÓDICA
1:200

KEYED NOTES:
NOTAS DE REFERENCIA:

1. CATHODIC PROTECTION TEST STATION
1. ESTACIÓN DE PRUEBAS DE PROTECCIÓN CATÓDICA.
2. INSULATED FLANGE KIT (IFK)
2. KIT DE BRIDA AISLADA (IFK)



DATE	APPR
DESCRIPTION	SYM
APPROVED	
FOR COMMANDER NAVFAC	
ACTIVITY	
SATISFACTORY TO DATE	
DES: SCW	CHK: LEI
FM/DM	CCA/EA
BRANCH MANAGER	
DESIGN MANAGER	
FIRE PROTECTION	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC BUREAU, VA	
DEPARTMENT OF THE NAVY	
NAVAL STATION ROTA	
REPLACE	
FUEL PIPELINE FROM PIER 3 TO BULK STORAGE	
CATHODIC PROTECTION PLAN	
PLANTA DE PROTECCIÓN CATÓDICA	
SCALE: 1:200	
EPROJECT NO.: 1606884	
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO. 14142836	
SHEET 135 OF 149	
EC110	

FILE NAME: U:\231423\04-Civil\01-SHEETS\UTIL-1606884-E-EC111.dwg LAYOUT NAME: EC111 PLOTTED: Tuesday, June 25, 2024 - 2:11pm USER: 6450

GENERAL NOTES:
NOTAS GENERALES:

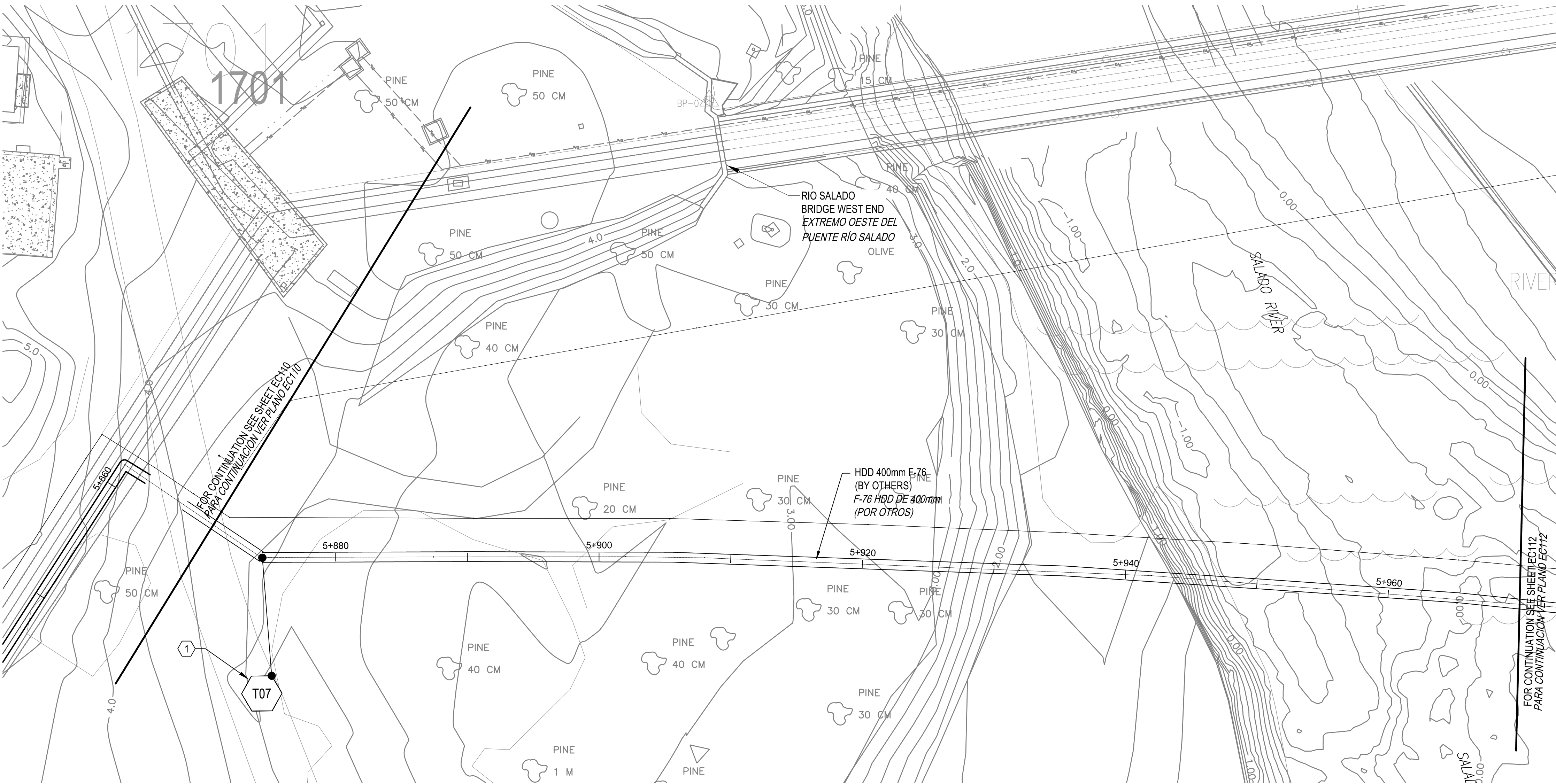
1. EACH TEST STATION HAS 0 ANODES CONNECTED.
1. CADA ESTACIÓN DE PRUEBA TIENE 0 ANODOS CONECTADOS.

B1

CATHODIC PROTECTION PLAN
PLANTA DE PROTECCIÓN CATÓDICA
1:200

KEYED NOTES:
NOTAS DE REFERENCIA:

1. CATHODIC PROTECTION TEST STATION
1. ESTACIÓN DE PRUEBAS DE PROTECCIÓN CATÓDICA.
2. INSULATED FLANGE KIT (IFK)
2. KIT DE BRIDA AISLADA (IFK)



DATE	APPR
DESCRIPTION	SYM
FOR COMMANDER NAVFAC	
ACTIVITY	
SATISFACTORY TO DATE	
DES: SCW	CHK: LEI
FM/DM	CCA/EAL
BRANCH MANAGER	
DESIGN MANAGER	
FIRE PROTECTION	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC	
NAVAL STATION ROTA	
REPLACE	
FUEL PIPELINE FROM PIER 3 TO BULK STORAGE	
CATHODIC PROTECTION PLAN	
PLANTA DE PROTECCIÓN CATÓDICA	
SCALE: 1:200	
PROJECT NO.: 1606884	
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO. 14142837	
SHEET 136 OF 149	
EC111	

FILE NAME: U:\23142\04-Civil\01-SHEETS\UTIL-1606884-E-EC115.dwg LAYOUT NAME: Layout1 PLOTTED: Tuesday, June 25, 2024 - 2:14pm USER: 6459

GENERAL NOTES:
NOTAS GENERALES:

- EACH TEST STATION HAS 0 ANODES CONNECTED.
- CADA ESTACIÓN DE PRUEBA TIENE 0 ANODOS CONECTADOS.

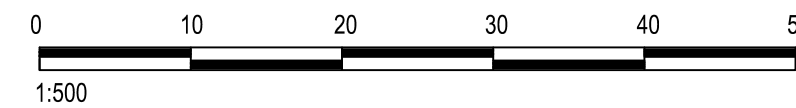
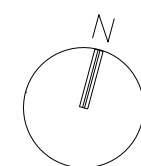
B1

CATHODIC PROTECTION PLAN
PLANTA DE PROTECCIÓN CATÓDICA

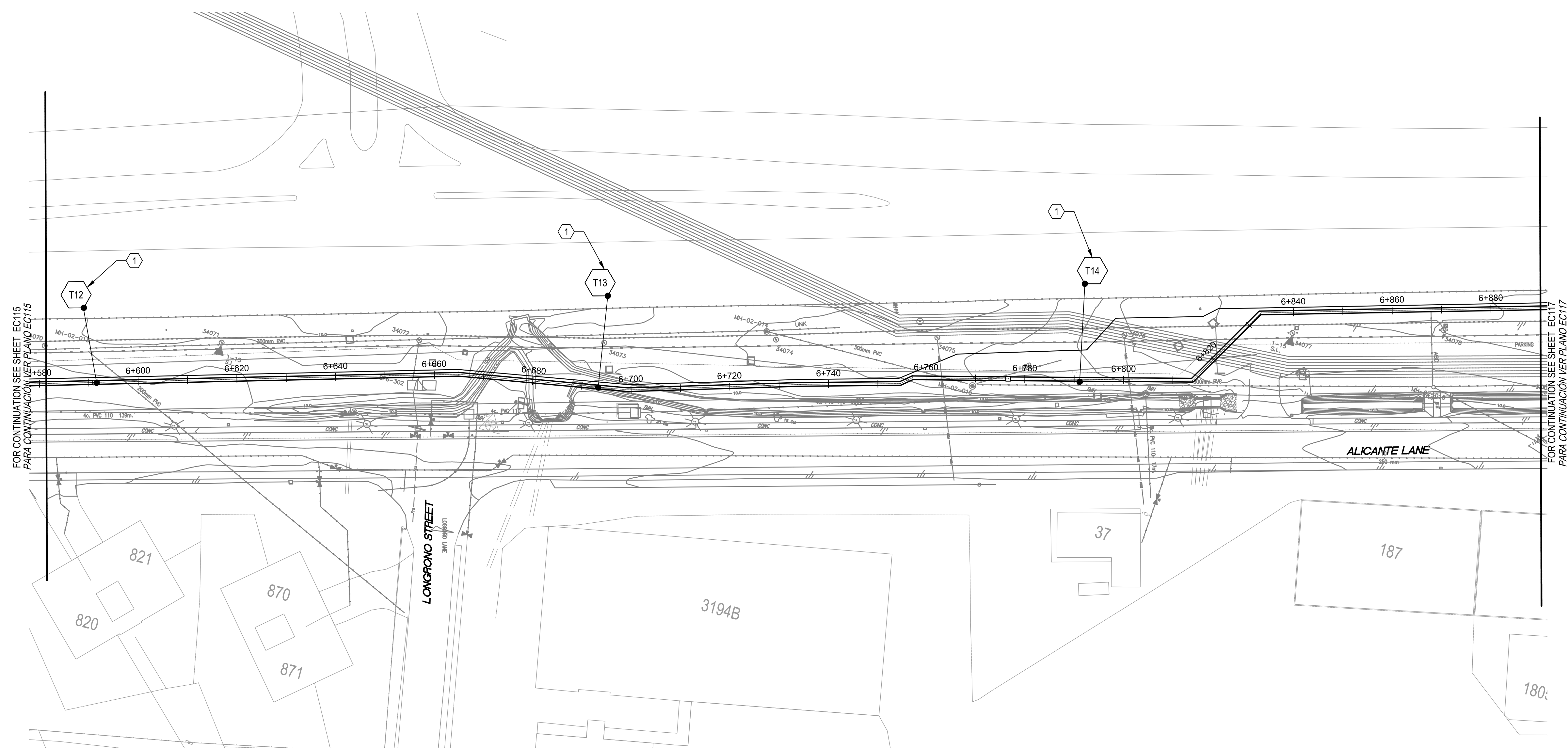
1:500

KEYED NOTES:
NOTAS DE REFERENCIA:

- CATHODIC PROTECTION TEST STATION
- ESTACIÓN DE PRUEBAS DE PROTECCIÓN CATÓDICA.



DATE	APPR
DESCRIPTION	SYM
FOR COMMANDER NAVFAC	
ACTIVITY	
SATISFACTORY TO DATE	
DESIGNER: SCW	CHECKER: GMR
PM/DM	CCA/EAL
BRANCH MANAGER	
DESIGN MANAGER	
FIRE PROTECTION	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC ATLANTIC DESIGN AND CONSTRUCTION	
REPLACE FUEL PIPELINE FROM PIER 3 TO BULK STORAGE	
CATHODIC PROTECTION PLAN PLANTA DE PROTECCIÓN CATÓDICA	
SCALE: 1:500	
PROJECT NO.: 1606884	
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	
14142841	
SHEET 140 OF 149	
EC115	



GENERAL NOTES:
NOTAS GENERALES:

1. EACH TEST STATION HAS 0 ANODES CONNECTED.
1. CADA ESTACIÓN DE PRUEBA TIENE 0 ÁNODOS CONECTADOS.

B1

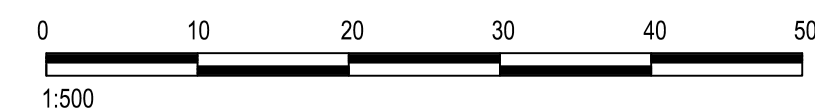
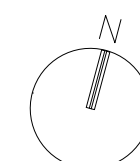
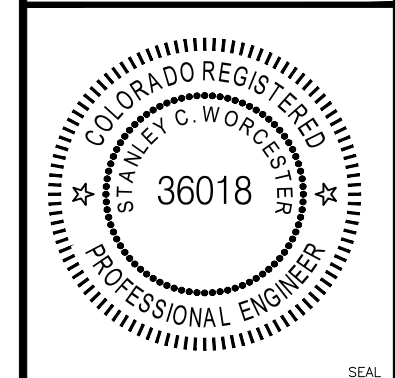
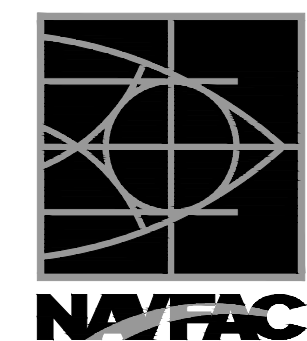
CATHODIC PROTECTION PLAN,
PLANTA DE PROTECCIÓN CATÓDICA

1:500

KEYED NOTES:

NOTAS DE REFERENCIA:

1. CATHODIC PROTECTION TEST STATION
1. ESTACIÓN DE PRUEBAS DE PROTECCIÓN CATÓDICA

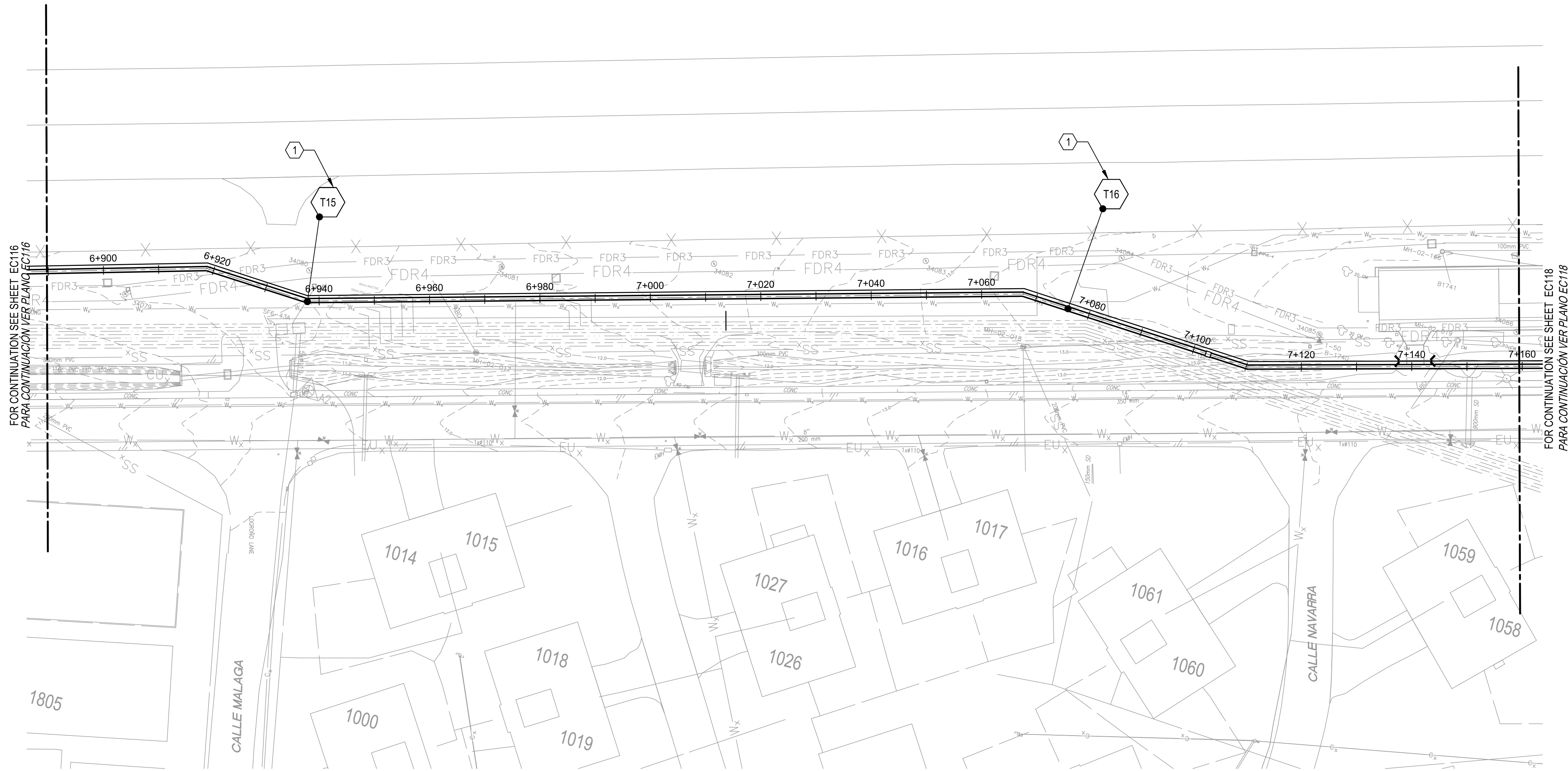
[illegible]

APPROVED _____			
FOR COMMANDER NAWAC _____			
ACTIVITY _____			
SATISFACTORY TO _____ DATE _____			
DES	SCW	DRW	GMR
PM/DM	CCA/EIL	CHK	LEI
BRANCH MANAGER _____			
DESIGN MANAGER _____			
FIRE PROTECTION _____			

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ATLANTIC DESIGN AND CONSTRUCTION	NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC NORFOLK, VA
NAVAL STATION ROTA REPLACE	ROTA, SPAIN
FUEL PIPELINE FROM PIER 3 TO BULK STORAGE	
CATHODIC PROTECTION PLAN <i>PLANTA DE PROTECCIÓN CÁTODICA</i>	

SCALE: 1:500	
EPROJCT NO.:	1606884
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	
14142842	
SHEET 141	OF 149
EC116	

FILE NAME: U:\231423\04-Civil\01-SHEETS\111-1606884-E-EC117.dwg LAYOUT NAME: EC107 PLOTTED: Tuesday, June 25, 2024 - 2:16pm USER: 6450



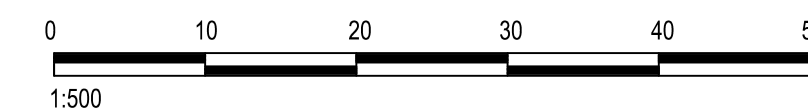
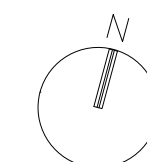
GENERAL NOTES:
NOTAS GENERALES:

- EACH TEST STATION HAS 0 ANODES CONNECTED.
- CADA ESTACIÓN DE PRUEBA TIENE 0 ÁNODOS CONECTADOS.

B1 CATHODIC PROTECTION PLAN
PLANTA DE PROTECCIÓN CATÓDICA
1:500

KEYED NOTES:
NOTAS DE REFERENCIA:

- CATHODIC PROTECTION TEST STATION
1. ESTACIÓN DE PRUEBAS DE PROTECCIÓN CATÓDICA.
- CATHODIC PROTECTION RECTIFIER NUMBER XX
2. RECTIFICADOR DE PROTECCIÓN CATÓDICA NÚMERO XX



DATE	APPR
DESCRIPTION	SYM
FOR COMMANDER NAVFAC	
ACTIVITY	
SATISFACTORY TO DATE	
DES: SCW	CHK: LEI
FM/DM	CCA/EA
BRANCH MANAGER	
DESIGN MANAGER	
FIRE PROTECTION	
DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC ATLANTIC DESIGN AND CONSTRUCTION	
NAVAL STATION ROTA REPLACE FUEL PIPELINE FROM PIER 3 TO BULK STORAGE	
CATHODIC PROTECTION PLAN PLANTA DE PROTECCIÓN CATÓDICA	
SCALE: 1:500	
EPROJECT NO.: 1606884	
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO. 14142843	
SHEET 142 OF 149	
EC117	

FILE NAME: U:\231423\04-Civil\01-SHEETS\UTIL-1606884-E-EC119.dwg PLOTTED: Tuesday, June 25, 2024 - 2:18pm USER: 6450 LAYOUT NAME: Layout1

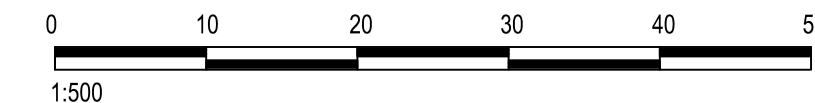
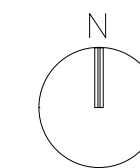
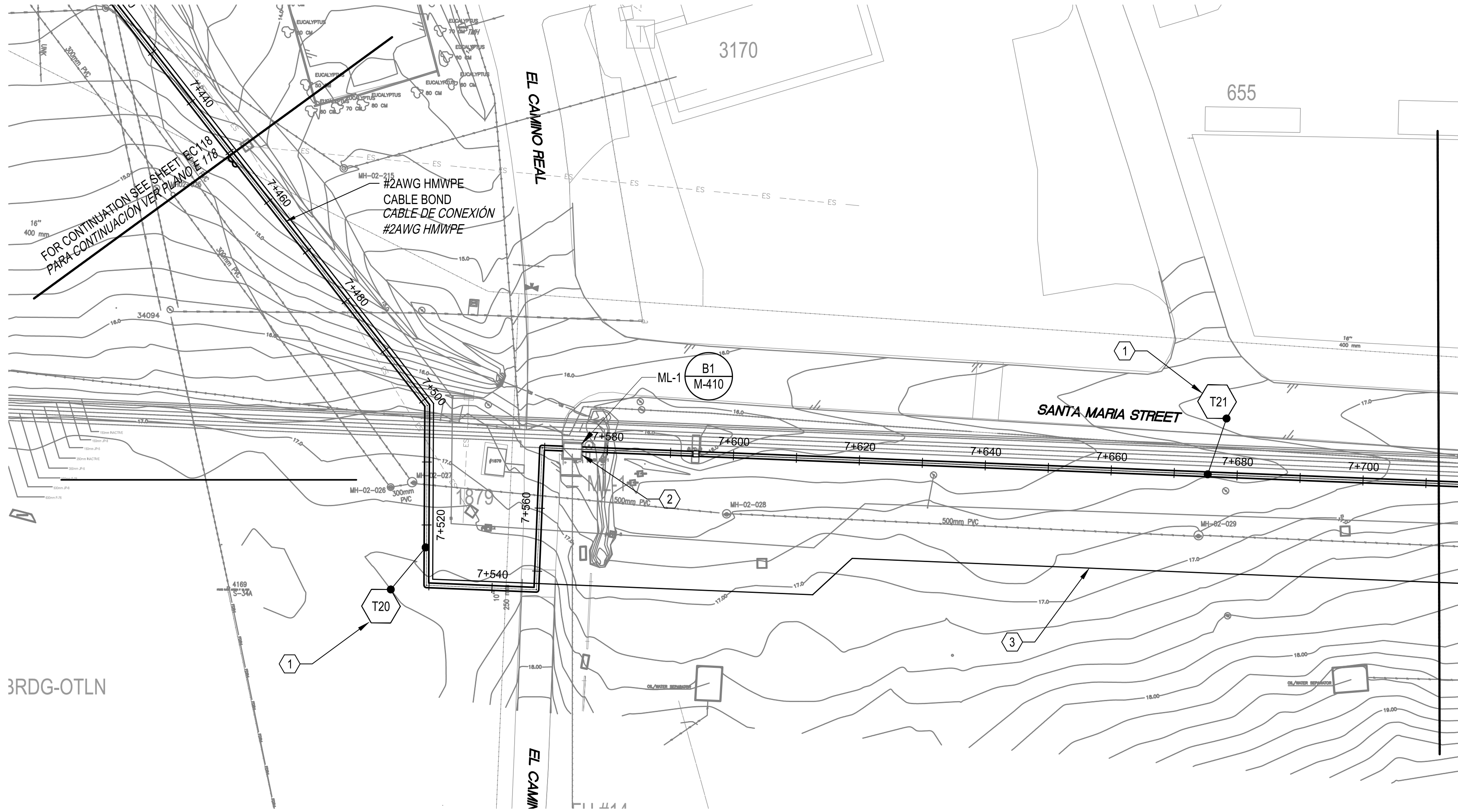
GENERAL NOTES:
NOTAS GENERALES:

- EACH TEST STATION HAS 0 ANODES CONNECTED.
- CADA ESTACIÓN DE PRUEBA TIENE 0 ANODOS CONECTADOS.

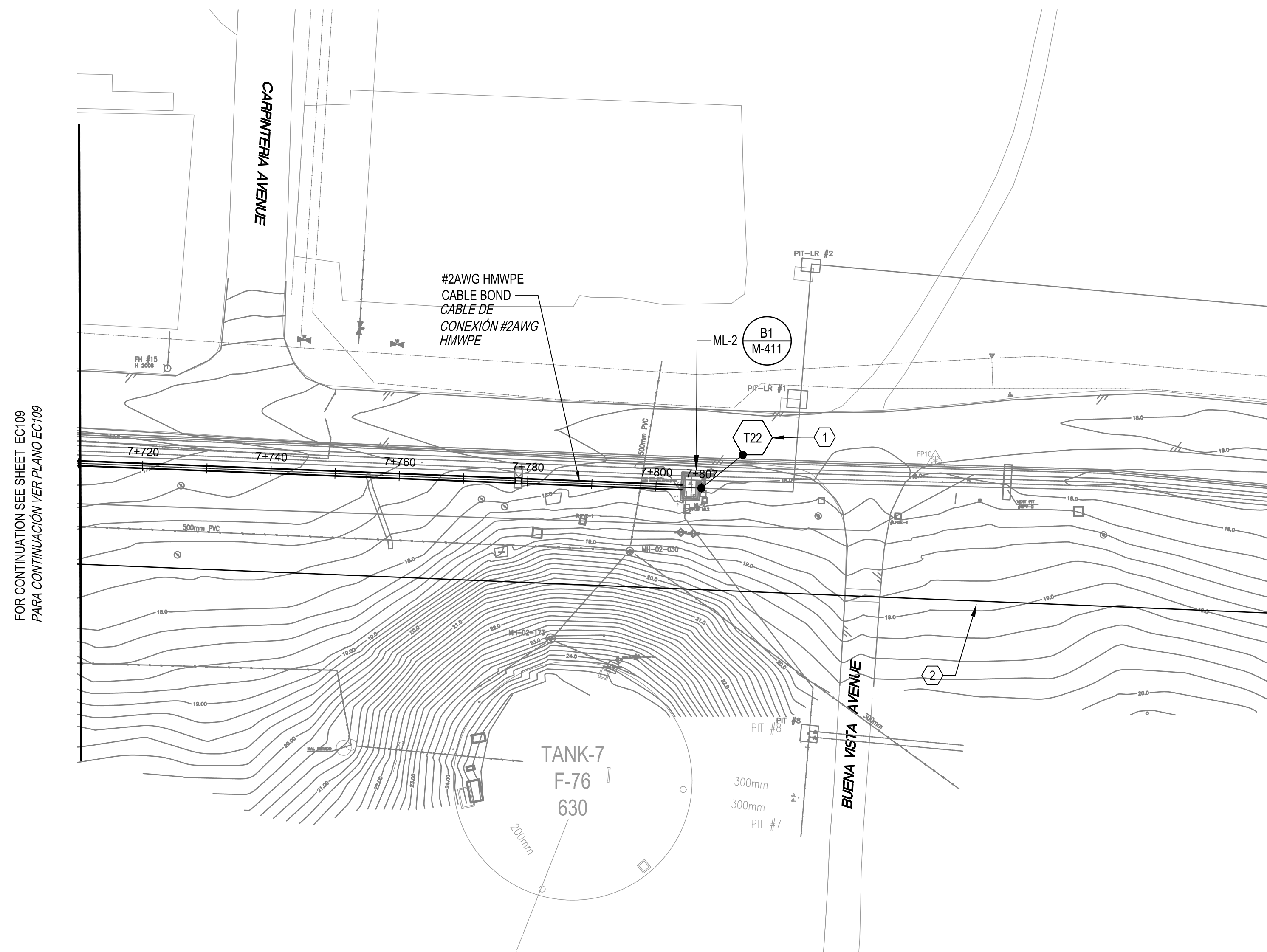
B1 CATHODIC PROTECTION PLAN.
PLANTA DE PROTECCIÓN CATÓDICA
1:500

KEYED NOTES:
NOTAS DE REFERENCIA:

- CATHODIC PROTECTION TEST STATION
ESTACIÓN DE PRUEBAS DE PROTECCIÓN CATÓDICA.
- CP TEST STATION AND BONDING BOX TO BE REMOVED DURING PIPE INSTALLATION AND REINSTALLED.
ESTACIÓN DE PRUEBA CP Y CAJA DE EMPALMES A RETIRAR DURANTE LA INSTALACION DE LA TUBERÍA Y REINSTALAR.
- 450 mm JP-5 LINE TO BE INSTALLED UNDER ANOTHER CONTRACT.
TUBERÍA JP-5 DE 450mm A SER INSTALADA BAJO OTRO CONTRATO.



DATE	APPR
DESCRIPTION	SYM
FOR COMMANDER NAVFAC	
ACTIVITY	
SATISFACTORY TO DATE	
DESIGNER: SCW	DRY: GMR
PM/DM: CCA/EA	CHK: LEI
BRANCH MANAGER	
DESIGN MANAGER	
FIRE PROTECTION	
NAVAL FACILITIES ENGINEERING SYSTEM COMMAND NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC ATLANTIC DESIGN AND CONSTRUCTION	
REPLACE FUEL PIPELINE FROM PIER 3 TO BULK STORAGE CATHODIC PROTECTION PLAN PLANTA DE PROTECCIÓN CATÓDICA	
SCALE: 1:500	
PROJECT NO.: 1606884	
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO. 14142845	
SHEET 144 OF 149	
EC119	



FOR CONTINUATION SEE SHEET EC109
PARA CONTINUACIÓN VER PLANO EC109

GENERAL NOTES:
NOTAS GENERALES:

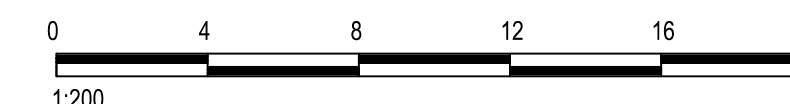
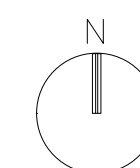
1. EACH TEST STATION HAS 0 ANODES CONNECTED.
1. CADA ESTACIÓN DE PRUEBA TIENE 0 ÁNODOS CONECTADOS.

B1 CATHODIC PROTECTION PLAN.
PLANTA DE PROTECCIÓN CATÓDICA
1:500

1:500

KEYED NOTES:
NOTAS DE REFERENCIA:

1. CATHODIC PROTECTION TEST STATION
1. ESTACIÓN DE PRUEBAS DE PROTECCIÓN CATÓDICA.
2. 450 mm JP-5 LINE TO BE INSTALLED UNDER ANOTHER CONTRACT.
2. TUBERÍA JP-5 DE 450mm A SER INSTALADA BAJO OTRO CONTRATO.

[illegible]

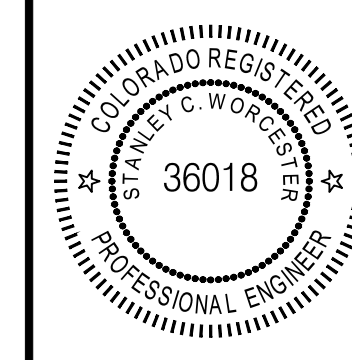
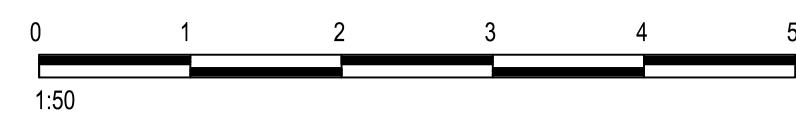
[illegible] N NAVFAC



A circle with a shaded sector. The arc length of the sector is labeled as 2.



- GENERAL NOTES:**
NOTAS GENERALES:
1. SEE GROUNDING DETAILS ON SHEET EG501.
1. VER DETALLES DE PUESTA A TIERRA EN PLANO EG501.



Stanley Consultants INC.
225 Iowa Avenue, Muscatine, Iowa 52761-376
www.stanleyconsultants.com

APPROVED			
FOR COMMANDER NAVFAC			
ACTIVITY			
SATISFACTORY TO		DATE	
DES	AT	DRW/GMR	CHK MZ
PM/DM	CCA/EJL		
BRANCH MANAGER			
DESIGN MANAGER			
FIRE PROTECTION			

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEM COMMAND NAUTIC DESIGN AND CONSTRUCTION NORFOLK, VA	NAVAL FACILITIES ENGINEERING SYSTEM COMMAND ~ ATLANTIC NORFOLK, VA
NAVAL STATION ROTA REPLACE	ROTA, SPAIN
FUEL PIPELINE FROM PIER 3 TO BULK STORAGE ELECTRICAL ENLARGED GROUNDING PLANS PLANTAS ELÉCTRICAS AMPLIADAS DE PUESTA A TIERRA	

SCALE:	1:50	
EPROJECT NO.:	1606884	
CONSTR. CONTR. NO.		
NAVFAC DRAWING NO.	14142849	
SHEET	148	OF 149
EG401		

1

2

3

4

5

D

C

B

A

ATTACH TO METAL FENCE POST WITH GROUND CLAMP
SUIETAR A POSTE METÁLICO DE VALLA CON PINZA
DE TIERRA

FENCE POST
POSTE DE VALLA

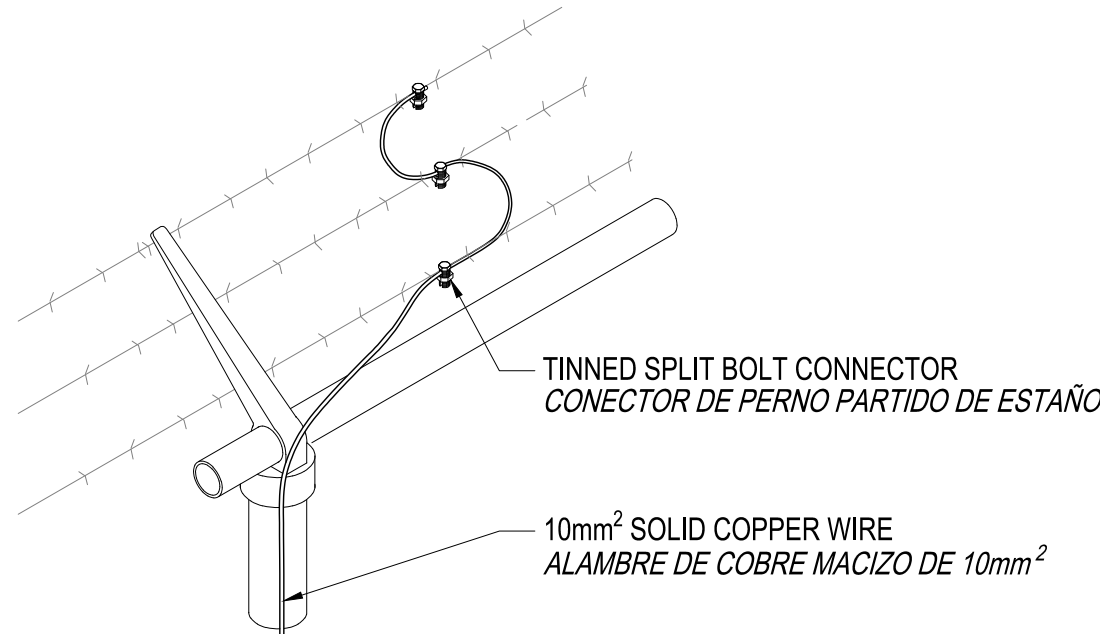
10mm² SOLID COPPER WIRE
ALAMBRE DE COBRE MACIZO DE 10mm²

MOLDED EXOTHERMIC WELD OR APPROVED
CLAMP-TYPE FITTING OF BRONZE
SOLDADURA EXOTÉRMICA MOLDEADA O
ACCESORIO HOMOLOGADO DE TIPO
ABRAZADERA DE BRONCE

COPPER-CLAD STEEL (19X3000)
GROUND ROD
PICA DE TIERRA DE ACERO (19X3000)
CON REVESTIMIENTO DE COBRE

C3

FENCE POST GROUNDING
PUESTA A TIERRA DE POSTE DE VALLA
NTS



TINNED SPLIT BOLT CONNECTOR
CONECTOR DE PERNO PARTIDO DE ESTAÑO

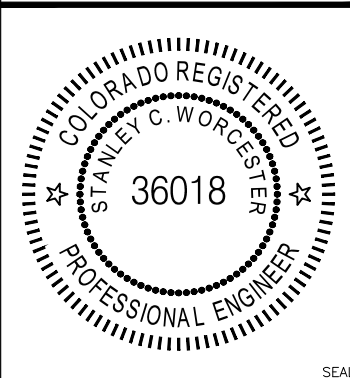
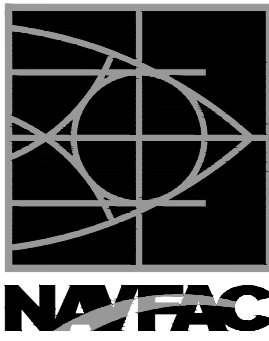
10mm² SOLID COPPER WIRE
ALAMBRE DE COBRE MACIZO DE 10mm²

C4

FENCE GROUND
TIERRA DE VALLA
NTS

KEYED NOTES:
NOTAS DE REFERENCIA:

- FENCE GROUNDING LOCATIONS ARE SHOWN ON DWG EG 401.
- UBICACIONES DE PUESTA A TIERRA DE VALLA SE MUESTRAN EN EL PLANO EG-401



225 Iowa Avenue, Muscatine, Iowa 52781-3764
www.stanleyconsultants.com

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DES AT DRW GMR CHK MZ

PM/DM CCA/EA

BRANCH MANAGER

DESIGN MANAGER

FIRE PROTECTION

NAVFAC DRAWING NO.

14142850

SHEET 149 OF 149

EG501

DEPARTMENT OF THE NAVY

NAVAL FACILITIES ENGINEERING SYSTEM COMMAND

ATLANTIC REGION

NAVAL STATION ROTA

REPLACE

FUEL PIPELINE FROM PIER 3 TO BULK STORAGE

ELECTRICAL GROUNDING DETAILS

DETALLES ELÉCTRICOS DE PUESTA A TIERRA

ROTA, SPAIN

WOLFOL, VA

NAVFAC DRAWING NO.

14142850

SHEET 149 OF 149

EG501

NAVFAC DRAWING NO.

14142850

SHEET 149 OF 149

EG501

NAVFAC DRAWING NO.

14142850

SHEET 149 OF 149

EG501

NAVFAC DRAWING NO.

14142850

SHEET 149 OF 149

EG501

NAVFAC DRAWING NO.

14142850

SHEET 149 OF 149

EG501

NAVFAC DRAWING NO.

14142850

SHEET 149 OF 149

EG501

NAVFAC DRAWING NO.

14142850

SHEET 149 OF 149

EG501

NAVFAC DRAWING NO.

14142850

SHEET 149 OF 149

EG501

NAVFAC METRIC DRAWING REVISION: 07 AUGUST 2018